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Remanent Magnetism of Late Secondary and Early Tertiary British Rocks†

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Abstract

The results of palaeomagnetic measurements are given both for the Tertiary basalts of North Eastern Ireland, and for the Cretaceous sediments in the south of England. The direction of magnetization of the basalts is consistently shallower than the present dipole field direction by about 8°. The Cretaceous results are shown to be much less likely to be reliable, but also indicate a shallower dip than that of the present dipole field. A simple and reliable statistical treatment is developed for use with palaeomagnetic data.

§ 1. Introduction

Rock magnetic data for rocks in Britain are completely lacking for the interval between Tertiary and Triassic times, except for a few Jurassic measurements made by Nairn (1937). Since the Triassic results (Clegg *et al.* 1954) show a marked variation from the present dipole field direction while the later Tertiary rocks are magnetized, on the average, in a direction corresponding to that of an axial dipole, it is extremely desirable to gather information concerning intermediate times, in order to ascertain whether any progressive change took place during the interval. The author's work has included two attempts to fill this gap in our knowledge; one was a survey of the Antrim Basalts of Northern Ireland, and the other a series of measurements on Cretaceous sediments from the Isle of Wight. These investigations are described in this paper.

§ 2. The Antrim Basalts

During a survey of the basalts of North Eastern Ireland, the author has measured the palaeomagnetic directions of several igneous bodies. This part of his work constitutes an extension of Hospers and Charlesworth's (1954) work in the same region.

The igneous rocks in this region are of Tertiary age, generally accepted as Eocene, but there exists some evidence, based on a comparison of Scottish and Irish Fossil pollens (Hospers and Charlesworth 1954), that they may have been formed as late as Miocene times. They are basaltic, and may be subdivided into three categories for the purposes of this report; (A) the older 'lower basalts' (olivine), (B) the younger 'middle basalts' (tholeiitic)

† Communicated by the Author.

which were previously called 'upper basalts' (Patterson 1955), and (C) the intrusive bodies, dykes and sills, whose relative ages cannot always be determined. Grouping of palaeomagnetic measurements has been made on this basis.

The principle underlying collection of samples was to collect as widely as possible geographically and temporally. The practical limitations of transportation often limited the collection to one sample per igneous body. Ninety samples were collected from fifty-seven distinct igneous bodies, at forty-seven different sites extending around the north-east coast of Ireland from Downhill, in the north-west corner of Co. Antrim, to the dykes exposed east of the Mourne Mountains. Collection was never made more than a few miles inland.

The measuring procedure was to cast each sample of average mass 6 kg, oriented in a rectangular plaster block, and then to measure its direction of magnetization by means of the astatic magnetometer built by Blackett (1952). A core $1\frac{3}{8}$ in. in diameter and $1\frac{3}{8}$ in. long was then cut from the block, and measured under a less sensitive magnetometer as a check on the first measurement.

Table 1 shows the three mean directions derived from the writer's measurements, and the equivalent data from Hospers and Charlesworth's (1954) paper. D is the magnetic declination measured in degrees east of North, I the inclination, $\mathcal{N}$ is the number of separate igneous bodies involved, N is the total number of samples taken from the $\mathcal{N}$ igneous bodies, and n is the total number of independent measurements made (plaster blocks and cores).

It has been customary among rock magnetists to use a form of statistics developed by Fisher (1953). These statistics are based on an assumed Gaussian distribution of density of measured vector directions about the mean direction. In certain cases this distribution is found experimentally to be invalid, so that statistics based on any assumed density distribution must be held in suspicion until the distribution be examined experimentally. In particular, any 'circle of confidence' containing the mean direction with a specified probability may contain falsely implied precision. For this reason a simple system of statistical evaluation of rock magnetic data has been developed (Appendix) and used on the data in this paper. The three significant quantities used are the mean direction, the analogue on a spherical surface of the standard deviation for a single reading, δ, and the standard deviation of the mean, ϵ. These can be found either for a set of igneous bodies, or for a set of measurements on a single igneous body. They are easily understood, and free from any assumptions about distribution.

The measurements for a single igneous body were rejected whenever δ exceeded 10°. The directions were corrected for geological tilt, which in general was small. In cases where only one sample has been taken ($N=1$, $n=2$) from an igneous body, δ has been redefined as the angle between the two directions measured for that sample.

It can be seen from table 1 that the four results are reasonably consistent, all reversed, and that all of them indicate a deviation from the present dipole field of about 8·4° towards shallower inclinations. This is a significant amount in terms of ϵ.

It is necessary to consider the possibility that this deviation might be due to a magnetostrictive effect caused by the pressure of the overlying lavas. The geology of the region indicates that the lowest of the basalts have not been buried more than about 600 metres deep. The pressure at this depth is not great, and similar pressures applied in the laboratory made no discernible directional change. There is no obvious discrepancy between the inclinations of the middle and lower basalts, even though they must have undergone quite different pressures during the geological time involved.

Table 1

	D(ETN)	I	$\mathcal{N}$	N	n	R	δ	ϵ
Lower basalts	173°	−64·6° up	19	35	70	17·599	22°	5·5°
Middle basalts	206°	−62·2° up	6	7	14	5·723	17°	7·1°
Intrusives	184°	−63·3° up	16	25	49	15·268	17°	4·4°
Above three combined	183°	−64·1° up	41	67	133	38·45	20°	3·1°
Hospers' results	194°	−60·2° up	24	72	120	23·253	14°	2·9°
Four results combined	187°	−62·4° up	65	139	253	61·63	18°	2·3°
Present dipole field	0°	+70·8° down	—	—	—	—	—	—
Present field	347°	+70·0° down	—	—	—	—	—	—

(For definitions of symbols, see text.)

On the assumption that the axis of the magnetic dipole field coincides on the average with the earth's rotational axis, the interpretation of this 8·4° deviation as being due to land-mass movement would indicate that Ireland has moved northwards 11·5° of latitude or about 800 miles since the eruption of the basalts.

§ 3. The Isle of Wight Cretaceous Sediments

There having been no igneous activity during Cretaceous times in Britain, a systematic attack was made on Cretaceous sediments from eight different localities in Southern England. Forty-six samples were collected from Compton Bay, Whitecliff Bay, Brixton Bay, and Sandown Bay on the Isle of Wight, and from Swanage, Durdle Door, Dungy Head, and Lulworth Cove along the Dorset coast. Each site presented cliff exposures.

Apparent magnetic stability was found in only nineteen of these samples, all of which were collected from the four miles of continuous Wealden sediment exposure along Compton and Brixton Bays. These nineteen

samples consisted of buff sandstones and red and grey clays. The other twenty-seven samples were rejected on the basis of the laboratory and geological tests mentioned below. From the nineteen samples, ninety-seven oriented 2 cm cubes were cut and their directions of magnetization determined. The mean intensities were 1×10^{-6} and 5×10^{-6} e.m.u./cm^3 for the sandstones and clays respectively. Table 2 gives the mean direction and standard deviation of the nineteen samples with respect both to the bedding planes and to the present horizontal plane. It also shows the

Table 2

	D(ETN)	I(down)	δ	ϵ
Present dipole field	0°	+67·6°	—	—
Present earth's field	351°	+64·0°	—	—
Mean direction with respect to horizontal plane	345°	+63°	4·9°	1·1° ($N=19$)
Mean direction with respect to bedding planes	2°	+59°	8·5°	2·0° ($N=19$)
Reverse of Eocene field direction (for Northern Ireland)	7°	+62°	18·5°	2·3° ($N=65$)
Triassic field direction (for Great Britain)	34°	+26°	—	—

directions of the present earth's field and the dipole field at the Isle of Wight, and the Eocene and Triassic field directions for Britain. Evidence concerning the magnetic stability of these nineteen samples was as follows:

(*a*) The mean direction of magnetization with respect to the present horizontal plane lies only 4° (3·5 standard deviations of the mean) from the direction of the present earth's field.

(*b*) The slight folding of these Wealden beds allowed us to calculate that the 8·5° standard deviation with respect to the bedding planes reduced to 4·9° with respect to the horizontal plane. This conforms with Graham's (1949) test of stability for folded strata.

(*c*) Alternating field demagnetization of the sandstones, in zero steady field, revealed a soft magnetic component, which disappeared for peak alternating fields less than 180 oersteds and had an initial magnitude which was small compared with a harder component. This harder component remained constant in magnitude and direction until 180 peak oersteds was reached, where it began to be unpredictably affected. Removal of the soft component improved the agreement between measurements on cubes from any one sample.

(*d*) Measurement of the magnetic moment of sandstone cubes suitably oriented in the earth's field revealed a soft component, which grew

exponentially with time, and was small in magnitude with respect to a harder component. The harder component was stable over at least several months.

(*e*) Two superficially different types of rock, with different natural intensities, comprised the nineteen samples.

The fact that the mean direction of magnetization is roughly intermediate between the Eocene and Triassic directions cannot be taken as proof of stability. Test (*a*) admits the possibility of the harder component possessing instability over a time greater than several months, but small with respect to the times involved in secular variations. It could, on the other hand, be a chance result of the particular tilt of these Wealden beds. Test (*b*) indicates instability of the harder component. Tests (*c*) and (*d*) reveal only that relatively soft and hard components do exist, and tell us nothing of magnetic stability over geological times, except that the alternating field test is consistent with the interpretation of the harder component as being thermoremanence in magnetite, following the work of Thellier and Rimbert (1955). Fact (*e*) is circumstantial evidence in favour of stability.

There is, therefore, doubt as to the validity of any palaeomagnetic interpretation of these results, and further work, preferably on similar sediments differently tilted, must be done to establish a conclusive result.

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APPENDIX

A statistical system has been used which employs the analogues, in the case of distribution of points on a spherical surface, of the standard deviations used for a distribution of points on a plane.

(*a*) The mean direction of N directions is found in the customary way. If λ, μ, ν be the direction cosines of the mean direction and λ_i, μ_i, ν_i the direction cosines of the N measured directions, then λ, μ, ν are defined as

$$\lambda = \sum\lambda_i/R, \quad \mu = \sum\mu_i/R, \quad \nu = \sum\nu_i/R$$

where $$R \equiv \sqrt{\{(\sum\lambda_i)^2 + (\sum\mu_i)^2 + (\sum\nu_i)^2\}}.$$

That is, the mean direction cosines define the mean direction.

(*b*) The mean cosine of the deviations δ_i from the mean direction is found in order to establish the standard deviation for a single reading.

$$\overline{\cos\delta} = \frac{\sum\cos\delta_i}{N} = \frac{\sum\lambda_i\lambda + \sum\mu_i\mu + \sum\nu_i\nu}{N} = \frac{R}{N}$$

whence the deviation $$\delta = \cos^{-1}(R/N).$$

This is the analogue of the 'standard' or r.m.s. deviation, since for small δ

$$\overline{\cos\delta} \doteqdot 1 - \tfrac{1}{2}\overline{\delta^2} = R/N.$$

This gives the r.m.s. deviation $\delta_{\text{r.m.s.}} \doteqdot \sqrt{\{2(N-R)/N\}}$.

(*c*) Then by analogy with the case of a planar distribution the standard deviation of the mean direction, ϵ, may be taken as

$$\epsilon = \delta/\sqrt{N}.$$

The expressions δ and ϵ are simple to use and understand, and express the facts adequately without recourse to assumptions about distribution of vector densities. They are also free of mathematical approximations which might limit their region of usefulness.